

# NGUYEN KHAC BAO

Fullstack AI Integration Intern

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## CAREER OBJECTIVE

AI integration intern candidate focused on turning AI models into usable products through backend APIs, RAG pipelines, vector databases, Docker, and cloud deployment. Hands-on with FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, and Cloud Run for deployable AI demos.

## EDUCATION

**Posts and Telecommunications Institute of Technology (PTIT)**

2022 – 2027

*Bachelor of Information Technology*

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

## TECHNICAL SKILLS

**AI Integration:** RAG pipelines, Gemini API, embeddings, prompt engineering, OpenAI-compatible APIs, LLM endpoint integration

**Backend / Cloud:** FastAPI, REST API, WebSocket, Docker, Google Cloud Run, Cloud SQL, Secret Manager, AWS EC2, Vast.ai GPU

**Databases:** Qdrant Cloud, MySQL/Cloud SQL, SQL basics, JSON/CSV data processing

**Frontend:** Streamlit, browser webcam demo, basic HTML/JavaScript

**ML / CV:** PyTorch, YOLOv8n, OpenCV, OCR, multi-object tracking, dataset QA

**Tools:** Git/GitHub, Linux, Cloud Build, Artifact Registry

## LANGUAGES

Vietnamese: Native | English: Technical reading proficiency for documentation, model cards, API references, and engineering articles.

## PROJECTS

**Vietnamese Legal RAG Chatbot**

2026

*Personal Project / Python, FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, Docker, Google Cloud Run / Source: [GitHub](#)*

- Built a cloud-ready fullstack RAG product demo with a Streamlit frontend and FastAPI backend deployed as separate Google Cloud Run services.
- Integrated Gemini as the default LLM and embedding provider; indexed the full 32,980-record Vietnamese legal QA dataset into Qdrant Cloud for retrieval-based answers.
- Used Cloud SQL/MySQL for chat history and environment-based configuration with Secret Manager for API keys and database credentials.
- Designed optional extensions for async document indexing with Celery/Redis-compatible broker and fine-tuned Vietnamese legal LLM serving through an OpenAI-compatible endpoint.

**Real-Time Face Mask Compliance Detection System**

2026

*Personal Project / Python, FastAPI, WebSocket, YOLOv8n, OpenCV, Docker, Google Cloud Run / Source: [GitHub](#)*

- Built and deployed an AI inference service with FastAPI REST + WebSocket endpoints and a browser webcam frontend running over Cloud Run HTTPS/WSS.
- Containerized the service for Cloud Run and logged warm endpoint latency around 151 ms for REST and 127 ms for WSS in the demo environment.
- Trained a YOLOv8n 3-class mask model on PWMFD converted from VOC to YOLO format: 9,205 images, 18,528 labeled instances, 0.970 mAP@50 validation.

**Traffic Violation Detection & License Plate Recognition**

2026

*Personal Project / Python, OpenCV, Ultralytics YOLO, ByteTrack, HyperLPR3, JSON / Source: [GitHub](#)*

- Built an AI video analytics pipeline for vehicle detection/tracking, traffic-light state estimation, stop-line event logic, plate OCR, and JSON evidence bundles.
- Prepared a compact CCPD2019 plate dataset with annotation parsing, YOLO bbox conversion, train/val/test split, negative samples, OCR crops, and QA reports.
- Fine-tuned a YOLO plate detector on 8,598 images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50 on validation.

**Autofall Labeler Tool**

2026

*Additional Project / Python, Streamlit, OpenCV, YOLO-Pose, Pandas / Source: [GitHub](#)*

- Built a human-in-the-loop dataset assistant with Streamlit review UI, metadata/annotation management, YOLO-Pose keypoints, QA reports, and reviewed-only export.
- Designed CSV-based source-of-truth records for annotations, metadata, keypoints, and review events to handle noisy CCTV data and reproducible dataset versioning.
- Achieved 90.2% coarse fall/non-fall agreement on a reviewed benchmark while identifying exact action-label accuracy as a limitation of weak temporal modeling.

## ACTIVITIES

**AI Research Group – University Club**

2025 – Present

*Member*

Discussed Machine Learning, Deep Learning, Computer Vision, and AI application development.